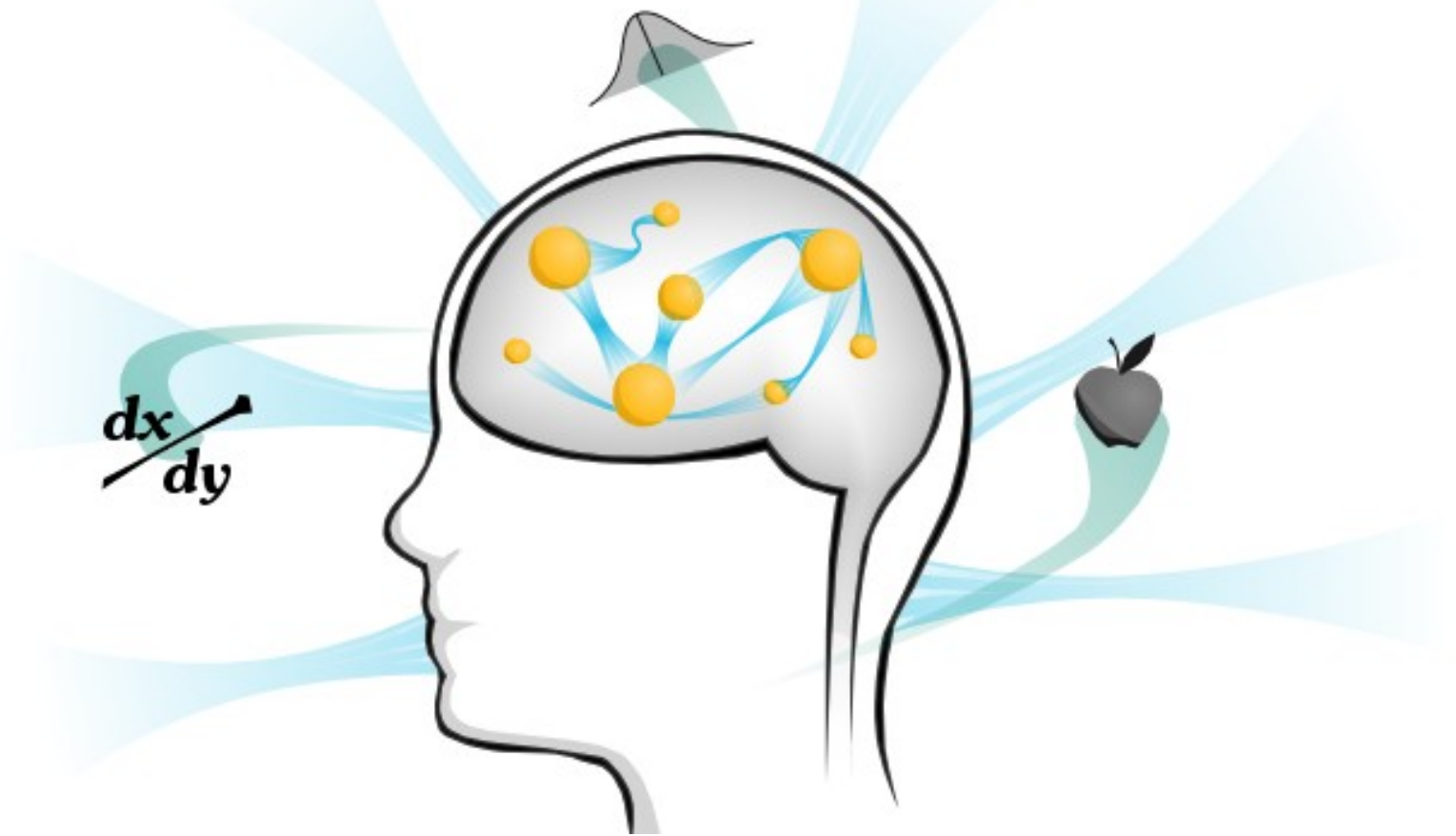


Learn More,
Study Less!



“If you understand something in only one way, then you don't really understand it at all. The secret of what anything means to us depends on how we've connected it to all other things we know. Well-connected representations let you turn ideas around in your mind, to envision things from many perspectives until you find one that works for you. And that's what we mean by thinking!”

-Marvin Minsky

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Introduction

What makes somebody smart?

Is it raw brainpower? Accumulated knowledge? Is it just academic learning or does it also include your experiences, people skills and intuition?

Intelligence is difficult to define. Although IQ tests and various exams try to measure it, deciding what makes somebody smart is hard to do. I prefer to avoid universal definitions and focus on a more practical one:

Being smart means being able to *learn quickly, remember a large amount of information* and be able to *sort that information in a way that achieves your goals*.

This is a much more personal definition. Learning goals can differ from person to person. You might just want to get A's in all your courses. John might want to become a master at computer programming. Susan might want to retain more from

the books she reads in her spare time. Another person might want to be able to apply concepts to business situations.

It's up to *you* to decide what being smart means.

This book isn't about reaching an arbitrary definition of intelligence. Rather, it's about giving you a strategy for learning. From this strategy, called holistic learning, I'll provide a set of different tools to put that theory into practice.

Some of these tools won't fit your learning style or goals. That's fine. My hopes is to provide a large assortment of tools that, with the backing of the holistic learning strategy, you can use to tailor your own approach. With practice many of these techniques can become powerful weapons in your learning arsenal.

This book has two major goals. First, the holistic learning strategy should give you a model for how to learn better. Without an easily understandable theory of learning, it is difficult to make improvements. By seeing the holistic learning strategy you have a basis for identifying your weaknesses and improving the way you learn.

The second objective of this book is to provide a variety of learning techniques. Throughout the book I'll explain what these techniques are and how they fit within the holistic learning strategy. Included with this book are exercise printouts so you can

practice these methods.

I wish you the best of luck in all your learning efforts and hope you enjoy the book. It took a great deal of effort, tweaking and experimentation to write. Hopefully that effort has been well invested and you can improve the way you think.

My Story

I've always been able to learn quickly. Getting A's and A+'s with little studying before tests wasn't a challenge for me throughout school. While in University, I've maintained an average that sits between A and A+. Despite this, I don't spend more than the average person on homework. In fact, I might even spend less.

Once, I wrote an inter-provincial test (I'm Canadian) for chemistry. The only problem: I didn't know I was supposed to write the test until a pencil and bubble sheet were sitting in front of me. On top of this, the test was on material I wasn't familiar with and topics that were never covered in my class. I was given an hour and a half to write the exam. I left after forty minutes because I wanted to eat lunch.

I won first place and received a check for \$400.

Self-learning has also occupied my time. I've taught myself several programming languages, business and writing skills and my bookshelf has hundreds of books I've read

in just the past two years. I've also dabbled in graphic design, musical composition and anything I could get my hands on.

Learning has always come easily to me.

Up until this point, I'd just be another smart kid. "Gifted" might fit as well, although there are people whose mental feats would put my small achievements to shame. I'd be just another kid who got a more favorable genetic cocktail, had pushy parents or some sort of glandular accident.

And if you read this far, you could probably slap on arrogant and boastful.

Until recently I probably would have agreed with you. But then something strange happened. I began to notice something different about myself and people even smarter than me. It wasn't just that smart people learned better or faster.

They learned *differently*.

Smarts requires a different strategy. Smart people had picked up different tactics,

sometimes intentionally but usually completely without awareness of them. It was these different strategies that made the difference in understanding.

That different strategy I called **holistic learning**. I call it holistic learning because it challenges you to view learning as a comprehensive whole, instead of a list of memorized facts. Smart people tend to make fewer distinctions between branches of knowledge and can easily relate one set of understandings to another.

By learning holistically, smart people are able to quickly integrate new information. More importantly, this information sticks. They actually “get” the concepts and see how the concepts relate to far more than just the problems given.

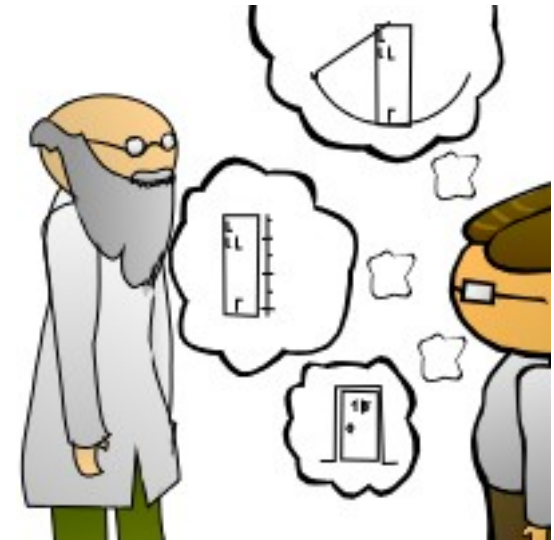
Once I was told a story that demonstrates this point perfectly:

Once upon a time, a student was in a physics class. He had achieved an otherwise perfect score, but the marker had graded him poorly on one question. The question had asked him how he would measure the height of a building using a barometer.

The student had written down, “Go to the top of the building. Drop the barometer and count the seconds until it smashes on the sidewalk below. Then use the formula for

acceleration by gravity to determine the height of the building.”

Of course, having referenced a barometer, the tester expected the student to use air pressure as a tool for measuring height. Since this answer did not demonstrate that the student knew how to solve questions about air pressure, he couldn't pass that portion of the test.



When the student brought up that his answer did solve the question being asked, the professor made a compromise. He said that he would let the student answer the question again with a different method. And if the student solved the problem again, he would award him the marks for the question.

Immediately the student responded that he would use the barometer to bang on the door of the landlord in the building. When the landlord answered the door, he would ask, “How tall is this building?”

At once, the professor saw what the student was doing. He asked him if he knew of any other methods to reach the answer. The student said that he did.

He recommended tying a long string to the barometer and measuring the length of the string. Or swinging the string as a pendulum and inferring the height by the motion it created.

The professor decided to award the student the marks. As the story goes, the student was a young Niels Bohr, later becoming the famous physicist and discovering the nature of electrons inside atoms.

This student didn't just know how to get the answer. He also understood the entire scope for which the problem existed. Instead of seeing the problem in the same terms he had been taught, he could easily view it a number of ways.

The goal of holistic learning is to replicate this process with the information you want to learn.

How to Use This Book

This book isn't designed to be read once, instantly making you a superior learner. Instead, think of it as a handbook. While the broader concepts of holistic learning might be understood the in the first reading, all techniques require practice. Expecting to master these techniques immediately is like trying to hit a bulls-eye with an arrow, when you've never held a bow before.

That is why, I've included with this book supplemental exercises and printouts. They can provide starting points the types of learning challenges these techniques are meant to solve.

Initially adopting these techniques may slow your learning down. Any transition to a new method will have an adjustment period. But once you become comfortable with these techniques you can learn more effectively without wasting as much time trying to relearn material you didn't understand the first time.

These techniques can't rest just within an academic setting. Holistic learning needs to encompass anything you want to understand. The more broadly you can apply these

methods, the better they function as a tool for increasing your understanding.

So the next time you pick up a book, attend a class or learn something new, think about the strategy of holistic learning. Ask yourself how you can apply it and what techniques you want to use. As the techniques become habits, it will be easier to permanently integrate any new idea.



Quick Tip!

Holistic learning introduces many new concepts and techniques. You'll get the best results if you focus on learning only one skill at a time. Throughout the second half of the book, you will see Mind Challenges. These challenges are set up specifically to help you adopt these new learning methods.

◦ Part I ◦

The Holistic Learning Strategy

What is Holistic Learning?

Holistic learning is the *opposite* of **rote memorization**.

Rote memorization involves repeating information enough times with the hopes that it will stick. Trying to remember a physics formula by repeating it to yourself dozens of times is learning by rote. This also a poor way to learn.

If you read the story about the student from the last section, you can probably see that smart people don't learn by rote. Do you think Niels Borh, as a young physics student, had formulas memorized in his head? Coming up with so many unique ways to solve a physics problem, it was the opposite. He understood what every symbol in the formula meant, and knew why it was there. He knew the rules so he knew how to break them.

Holistic learning is a theory for learning that more accurately describes how your brain works. Your brain isn't the same as a computer filing system. Computer files are stored in strips of 1's and 0's in locations on a hard drive. Your brain stores information

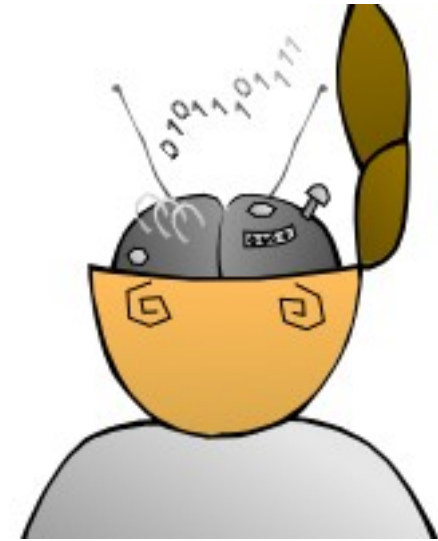
as associations between billions of neurons.

Rote memorization might work if we had computer brains. All it would take is an accurate copy of information and you could memorize anything. Unfortunately, we don't have computer brains and that is why rote memorization is a less effective way to learn.

Holistic learning takes a different approach. Instead of trying to memorize information by making a perfect copy in your brain, it uses the web of neurons you have. Holistic learning creates webs of information. One idea relates to another idea. That interrelating of ideas allows you to easily navigate through complete understandings.

With holistic learning, ideas aren't learned in isolation. If you follow holistic learning closely enough, you'll soon realize that it is impossible to learn an idea in isolation. Learning anything requires associations. The more associations you can create and the stronger those associations are, the better.

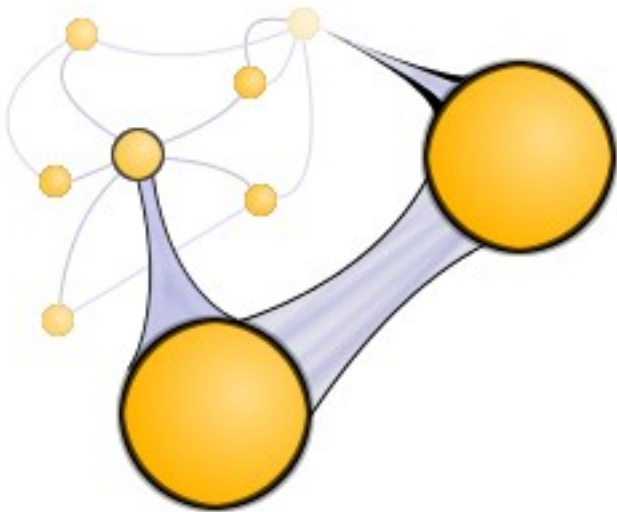
Rote memorization would suggest learning fits into neat little boxes. A box for math containing algebra, arithmetic and calculus. Your calculus box contains more



boxes for how to find the derivative of a function, and a few common applications of those rules. Nowhere does your calculus box contain information on biology, history or the plot of a science fiction movie.

But learning doesn't fit into boxes. Learning fits into webs.

A holistic learner wouldn't take such an organized approach to storing information (which might explain why so many smart people have horrible organization skills). Instead holistic learners connect everything. A derivative isn't just a formula, it's a feeling, an image and you can relate it to flying a supersonic jet.



Rote memorization seems to make sense when you can't see the alternative. When you don't know the steps learning should follow and lack techniques to move through those steps, simply pounding information in your skull seems to work. This is like a caveman using a rock to hunt mammoths instead of a rifle. Until you understand the steps and techniques for learning, rote memorization is

crude and inefficient, but it still works better than nothing.

How You Store Information

Holistic learning is my hypothesis for how learning actually works. I say hypothesis because holistic learning is less scientific fact and more practical metaphor. Science still has a long way to go to discover the physical connections in the mind. Holistic learning simply suggests one way of viewing how smart people manage to learn.

Whether electrons actually exist as billiard balls or violin strings jumping around a cloud of protons is less important than the implications of these metaphors. Holistic learning, similarly, is about providing an easily accessible theory that seems to fit how information is stored, rather than a description of the biological processes buried in your mind.

Holistic learning is based on three main ideas:

- 1) Constructs
- 2) Models
- 3) Highways

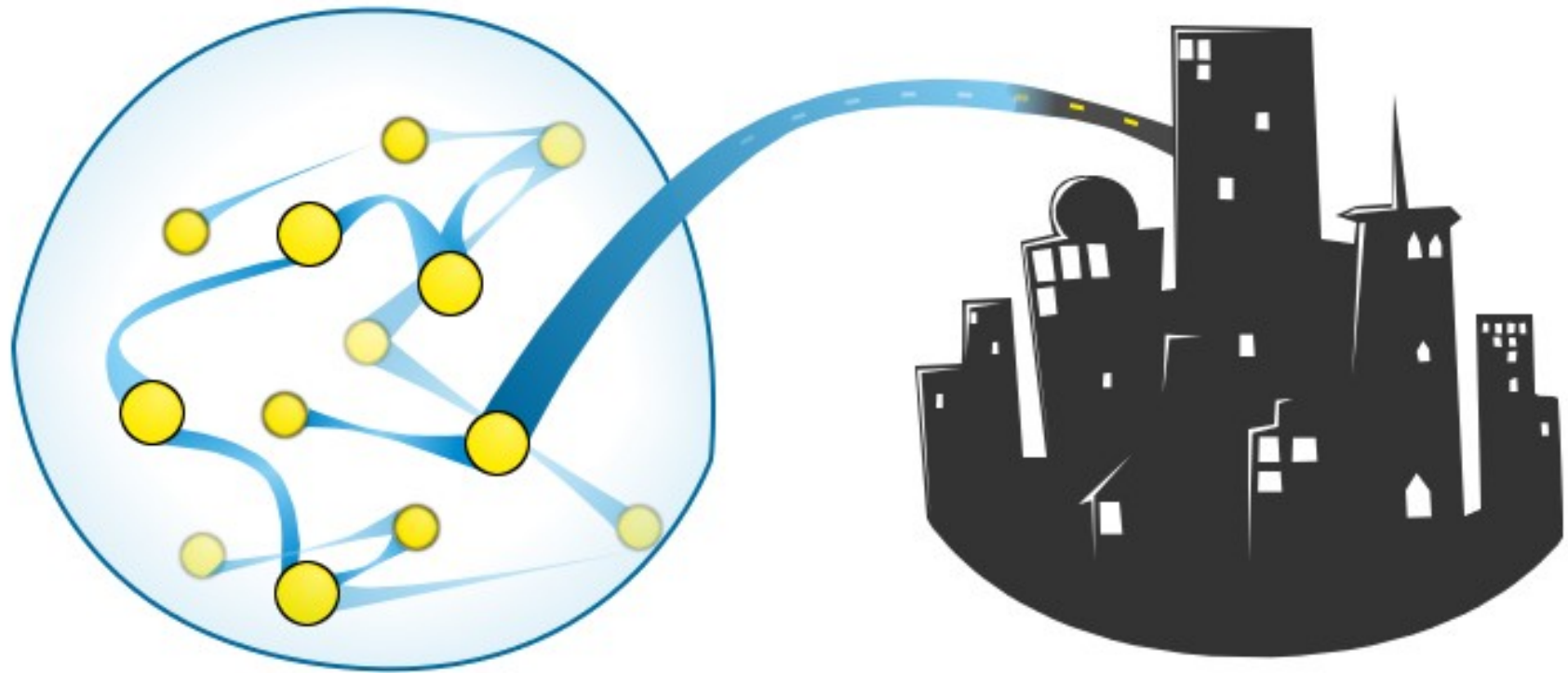
As I'll explain, seeing how these three elements fit together makes applying the holistic learning strategy easier. Just as knowing the different chess plays is useless without knowing that a bishop moves diagonally, knowing the holistic learning strategy is useless unless we both operate from the same points of reference.



Quick Tip!

Holistic learning is based on linking ideas as a way of remembering and applying them. The best techniques to start learning are metaphor, visceralization, flow-based notetaking and diagramming. These methods form the foundation of holistic learning.

Constructs



Constructs

A construct defines a set of tightly interlinked understandings. For example, if you've tried your hand at learning to create programs with C++, you probably have a C++ construct. This construct defines all your knowledge about C++ and is the sum total of all the interconnected ideas about C++.

Think of a construct as being like a city in your mind. Within a city you have thousands of buildings you can travel between. Some are big and important and are linked by roads to hundreds of other buildings in your city. Others are less important and only have a few dirt paths leading to them.

In the landscape of your knowledge, your mind is full of these cities. Your basic math and language skills are probably large, efficient cities. You have no problem navigating from $3+4-10$ to $10*3+7$. These questions are easy because your constructs for basic math are highly developed.

Understanding is the result of a highly developed construct.

Are there certain subjects you just “get”? These subjects are easy for you to understand and seem obvious. Chances are this is because those subjects lie within

highly developed constructs. They are like cities where the roads are clean and free of traffic congestions. The buildings are maintained and rarely are entire pathways blocked off.

In contrast, are there certain subjects you have difficulty understanding. These would be like newly constructed or poorly designed cities. The roads don't lead to many places. It is easy to get lost and there are many buildings which need to exist but either haven't been built yet or have been lost in the maze of roads and wrong directions.

Ask yourself this question: In a city which building would be the easiest to locate?

- A) One that is connected to many other buildings through hundreds of roads, or
- B) One that has a single dirt path connecting it to only one other building.

The building in A would be much easier to find. If you got going down the wrong path, you wouldn't have much trouble eventually reaching your destination. With the situation in B, you are relying on chance that you stumble upon the only building that connects with your final location.

When building constructs, your goal is to create as many possible interconnections as possible between ideas. Smart people tend to do this process naturally. As each concept comes up, it is automatically linked with other ideas.

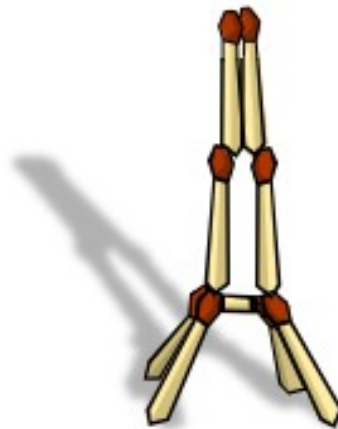
You probably use this process already, but perhaps not as effectively as people who learn effortlessly. The holistic learning strategy has many different techniques for manually linking ideas. Once those methods become habits, it doesn't take long before new connections are made.



Quick Tip!

Your constructs weren't built in a day. There is no magical technique to immediately “get” any subject. Having a powerful construct comes from linking concepts together, one idea at a time.

Models



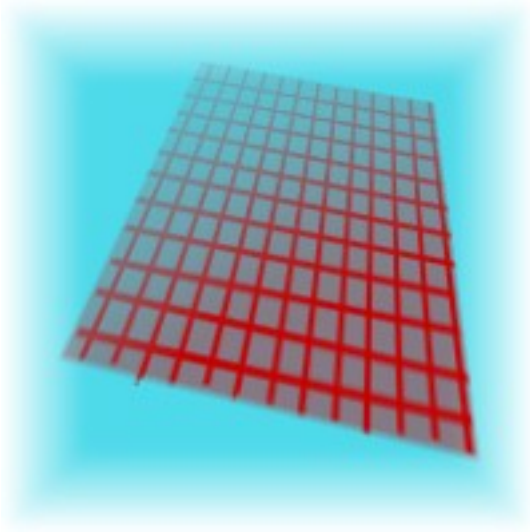
Models

If constructs are the door to holistic learning, then models are the key. Models are simplified constructs. They are snapshots of larger understandings designed to be portable and easy to store. Models are essential when trying to learn new concepts quickly.

To understand what a model is, think about this book. A model for this book would be the table of contents. In a few pages, they summarize the thousands of words contained in the book. If you wanted to describe the contents of this book quickly, you would probably start there instead of reading the entire book word-for-word.

Models can take a variety of forms. The goal, however, is always the same: compressing information. By taking several core concepts and linking them together into one form, you create a model. Models are essential when you are just starting to build a construct or making large additions to current constructs.

I used a model when I was first learning what a subspace was. A subspace is a mathematical term defining a portion of a larger vector space. An example of a vector space might be three dimensions. A subspace within that vector space could be a two dimensional plane.



The model I used to represent this subspace was a bright, blue infinite background. This was the three dimensional vector space. Then I visualized a red translucent grid slicing through the background. This was my subspace.

This model only represents some possibilities for subspaces. A subspace could also be a line within a plane, or a section of an entire matrix of numbers. Models do not need to have perfect accuracy, they only need to combine several ideas to make them more manageable. As my understanding of subspaces grew, I made adjustments to my previous model and created new models. Once I had a construct, I could build out that understanding and correct any errors.

Another time I used a model was when teaching myself computer programming. Programming languages make use of a concept known as a variable. Variables are used to store information that might change throughout the running of a program. A username, counter or key could all be stored with variables.

I created a clear image of the variables as being types of jars, and the concept made sense. Since most variables are broken into various types (some store numbers, others

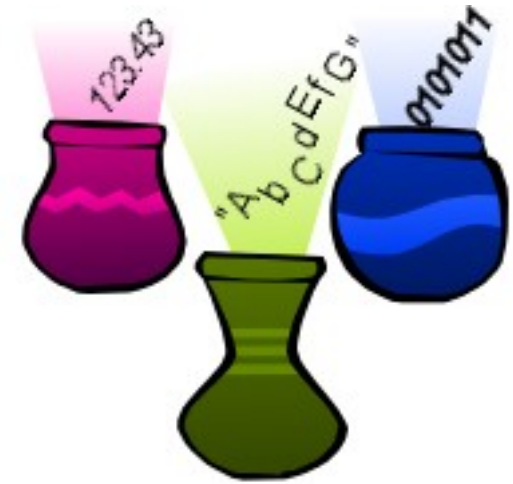
words or letters) I could imagine different jars with different openings for holding different types of data.

This model for variables stored several key ideas into one image or concept. This compression formed the seed for which future understandings could be linked and refined.

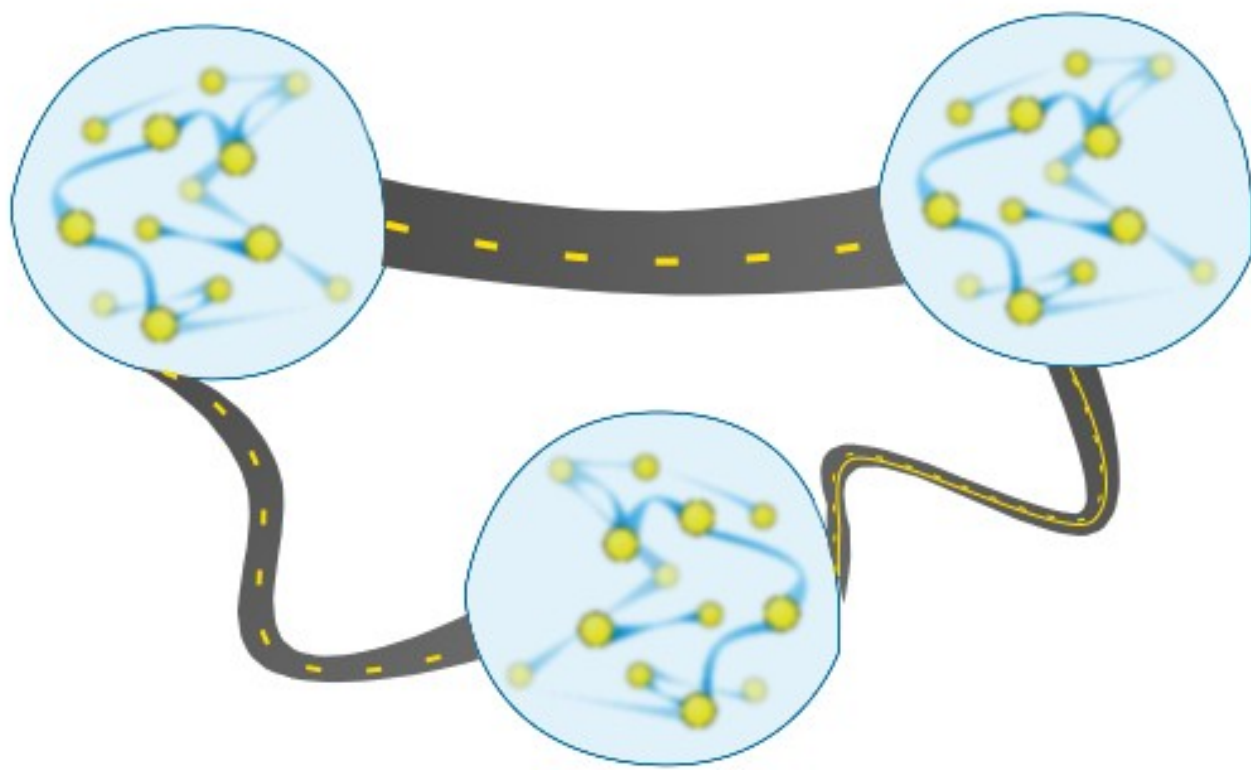
A model doesn't need to be an image. It only needs to be an existing understanding that compresses several key ideas into one. There are many ways to generate and expand models which I'll discuss later in the book. Visualization is only one method.

When I learned about functions in computer programming, I imagined them working like a pencil sharpener. Dull pencils go in, and a casing concealing the machinery sharpens the pencil. Sharp pencils come out. This model doesn't need to rest on a specific image, but just the general concept of how a pencil sharpener works. One construct lending tools to build models for another.

The takeaway point about models is that they act like seeds for constructs. Like a small group of pioneers with compact building materials heading to fresh ground to start a new settlement. A compressed set of key ideas that can be expanded.



Highways



Highways

Constructs are built on roads linking buildings with one another. While having hundreds of intercity roads is useful, it doesn't make use of the real power of your holistic brain. In order to do that, you need to add highways: linkages between isolated constructs.

A highway is a reference that links two completely different ideas. If you were reading a biology textbook and made comparisons from evolution to business courses you are taking, that would be a highway.

The benefit of a highways isn't an immediately obvious one. If you want to succeed in school, most instructors will emphasize having well defined constructs, not having a brain full of highways.

However, highways help with creativity. Thinking "outside the box" might as well describe people who think outside of constructs. Highways allow them to do this by making connections between areas that they didn't previously think were connected. Creativity can be seen as being able to utilize these highways to build new constructs in territories that were previously empty.

An example of when I've used a highway before was when I was reading *The Prince*, by Nicolo Machiavelli. This book outlines Machiavelli's principles for governing

a state, ruling people and conquering other nations. Today it's largely seen as either a work of a devious genius or a psychopath.

When learning about the ideas Machiavelli presents in his book, I needed a basis to compare them to. As a result, I created my own highways between Machiavelli's advice on statecraft and my personal experience with relationships, social settings and management. Although some of his ideas may have flaws, this approach allowed me to completely understand the ideas before deciding which I would accept or reject.

Building highways offers you more flexibility as a learner. Normally, if you don't understand the way an instructor teaches a subject, you're completely lost. With highways, you can import your own understandings from different subjects.

Familiar Constructs

There are several constructs most people possess in a highly developed form. These familiar constructs are so ingrained, that they serve as an invaluable resource for models, highways and learning new subjects. There are probably many more you possess, but here are a few likely candidates:

Sensory Constructs

These are probably your best developed sets of understandings. You have hundreds of images, sounds and feelings linked together through experience. You might even want to consider sensations to be the root construct that all others are built on.





Relationship Constructs

Stories, people and interactions define most of our existence. Is it any wonder then that stories make new subjects easier to learn? Saying that halogen compounds have one less electron than noble gases is different than saying they're neighbors on the periodic table.

Basic Math Constructs

Depending on your math background, you may find anything from arithmetic to advanced calculus to be painfully obvious. These can be used as the basis for building further math constructs or you can use them to simplify linkages in other subjects.

$$2 + 2 = 4$$



$$ax^2 + bx + c = 0$$



$$\log_3()$$

I'll reference these and other familiar constructs when offering suggestions for implementing various techniques. Think of these as your capital cities which smaller settlements can be modeled after.

Constructs, Models and Highways, Oh My!

All of this talk about constructs and models may seem abstract to start with. Don't worry. The reason I bring them up is because they form the backbone of holistic learning.

Many books strive to teach different learning techniques. These are similar to the techniques I'll describe. The difference with holistic learning is you have a background theory which explains why these techniques should work, and give you a better sense for modifying and creating your own.

By providing the background of how information is stored in your brain, hopefully it will lead you to adopt approaches that make sense within this strategy. You wouldn't try to save computer files by telling the computer a story or typing them out twenty times. That doesn't make sense given how we understand computers store information. Understand how your brain stores information and you can enhance the way you learn.

The Sequence of Holistic Learning

Up until this point I've described what information looks like when it is stored in your head. But the real question is, *how does it get there in the first place?*

I've divided up almost all learning tasks into several unique steps. Often you won't need to complete all the steps to understand enough to pass a test. The steps might not always follow in the same order. You may go back to the first step after completing the second before moving onto the third.

What is important is that each of these steps represents key activities involved in learning. When you repeatedly miss a step in the process, or finish it poorly, the information won't be stored properly. Learning fails and the information doesn't make it into your holistic web.

By examining each step and seeing how it fits into our holistic learning theory, we can have a strategy for learning. With this strategy, you can use specific tactics to fix holes in your current technique. You can also develop new methods to replace inefficient ones you have for various steps along the holistic learning process.

The sequence of holistic learning is:

- 1) Acquire
- 2) Understand
- 3) Explore
- 4) Debug
- 5) Apply

A final step which exists outside the sequence but applies to every step is Test. Testing is checking to see how well you are executing each of these steps. Without rigorous self-testing (by the time you reach the exam room, you're too late) you may not realize a step was poorly executed.

Here are the six steps:

- 1) **Acquire** - The point at which information enters through your eyes and ears. Reading, taking notes in a class or personal experiences are all part of the Acquire Phase. The goal here is to get accurate information in the most compressed form.
- 2) **Understand** - Understanding means taking raw information and giving it a context. This would be the most basic interlinking you would need to perform in order to learn.
- 3) **Explore** - The Explore Phase is really where holistic learning takes full force. Here you form the models, highways and broader connections needed for well defined constructs.
- 4) **Debug** - The Debug Phase looks for errors in your models and highways. This phase prunes back your connections so invalid ones won't remain, or will be constrained to the area they work.
- 5) **Apply** - The Apply Phase takes debugging to the final level by making adjustments compared to how this information operates in reality. Having a set of understandings is useless if they aren't tailored to the real world. Failure on this step could be an example of people who have book-smarts but can't seem to use them outside the class.

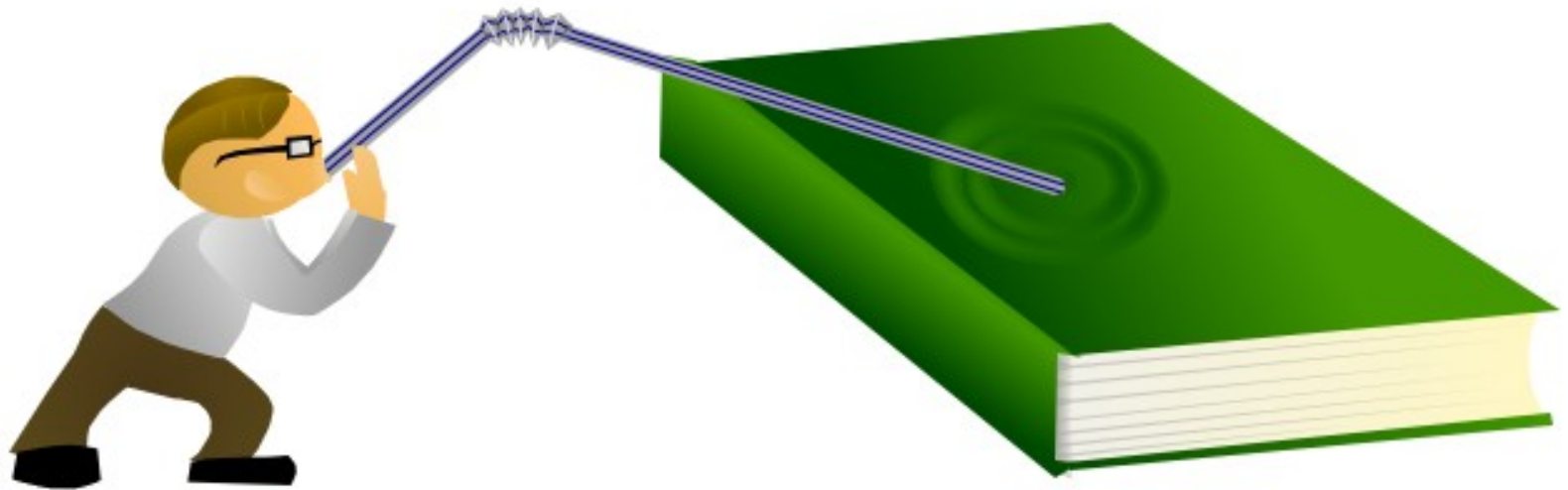
6) **Test** - Continuously you should be testing your methods and learning throughout all six stages. This will help you quickly identify problems in the system you use for learning, and help you develop new techniques to combat weaknesses.



Quick Tip!

The different techniques in the second half of this book are suited towards different stages in the sequence. Speed reading can help in acquiring ideas, flow-based notetaking and metaphor can help in exploring ideas. Model debugging and project-based learning can help you debug and apply ideas.

The Acquire Phase



The Acquire Phase

This phase represents the path information takes to enter your brain. Your ability to handle different formats of information and to eliminate clutter are essential here. Here are a few things to consider when evaluating how well you perform in this phase:

- How long does it take you to read a page of material with 90% comprehension?
- How much of your lectures do you actually pay attention to?

In all cases, your ability to handle a medium of information forms the first filter which can enhance or destroy your learning efforts. Any information lost here is gone forever. No amount of holistic learning techniques can make up for a failure to acquire the important ideas.

When acquiring information, you have three major goals:

- 1) Simplicity
- 2) Volume
- 3) Speed

Acquiring for Simplicity

Read the following words:

“I am. I am. I am. I am. I am. I am.”

What is the point of reading all these words? There is a lot of redundancy as many words are repeated. You could have simply read: “I am. x6” and that would have contained exactly the same information.

Reading requires that you consume the purest information possible. While most authors and lectures don't fill space with dead air, they will often give you information you don't need to form models and constructs. A professor that breaks down the meaning of a formula may be giving a lot of redundancy if you can determine how the formula is created on your own.

Simplicity means you acquire the information with the lowest amount of redundancy. Supporting examples are important to the degree that they help you form models. When you don't need them, don't bother acquiring them.

Acquiring for Volume

As long as you account for simplicity, you should be trying to get the most information possible. Reading one sentence from this book offers far less understanding than reading every page. The more information you can process, the more you will understand. Someone who reads a 100 books a year will usually have more knowledge than someone who reads only 2 or 3.

Acquiring for Speed

The final goal of acquiring information is speed. All things being equal, reading a book in thirty minutes is better than taking an hour. Speed tends to work against volume and simplicity. The faster you go, generally the less information you can absorb and the less discriminating you can be in determining what to acquire.

The Acquire Phase can be enhanced by adopting better reading and listening methods. Even at this early stage, most people have significant room for improvements. I'll discuss some of these methods in Part II of this book.

The Understand Phase



The Understand Phase

Input is nothing without comprehension. If you don't understand the surface of what a book is trying to say, the chances of remembering it for an exam or applying it in real life are almost nonexistent. Most people intuitively feel the Understand Phase. The information makes sense even if it still isn't firmly rooted into your brain.

The holistic learning sequence could be seen as taking three layers, each expanding the degree of understanding. The Understand Phase forms the first layer, followed by Explore and finally Apply. Each provides more depth to the information you come across.

With the Understand Phase, you are getting at the surface of information. As an example, say you are learning a new mathematical formula. You know the basics of what the symbols in the formula stand for, and possible problems that could be solved using this formula. If you repeat the formula enough times, you might even remember it. Learning by rote is learning with only the Understand Phase.

The Understand Phase, is where most people stop and smart people go beyond. Simply reaching this phase shows that you have linked the information enough so you can understand the context. The formula isn't just random squiggles and letters.

Unfortunately, it isn't much more. You might not be able to determine how the formula was derived, how it relates to other formulas or how it can be applied to problems outside what you've been taught.

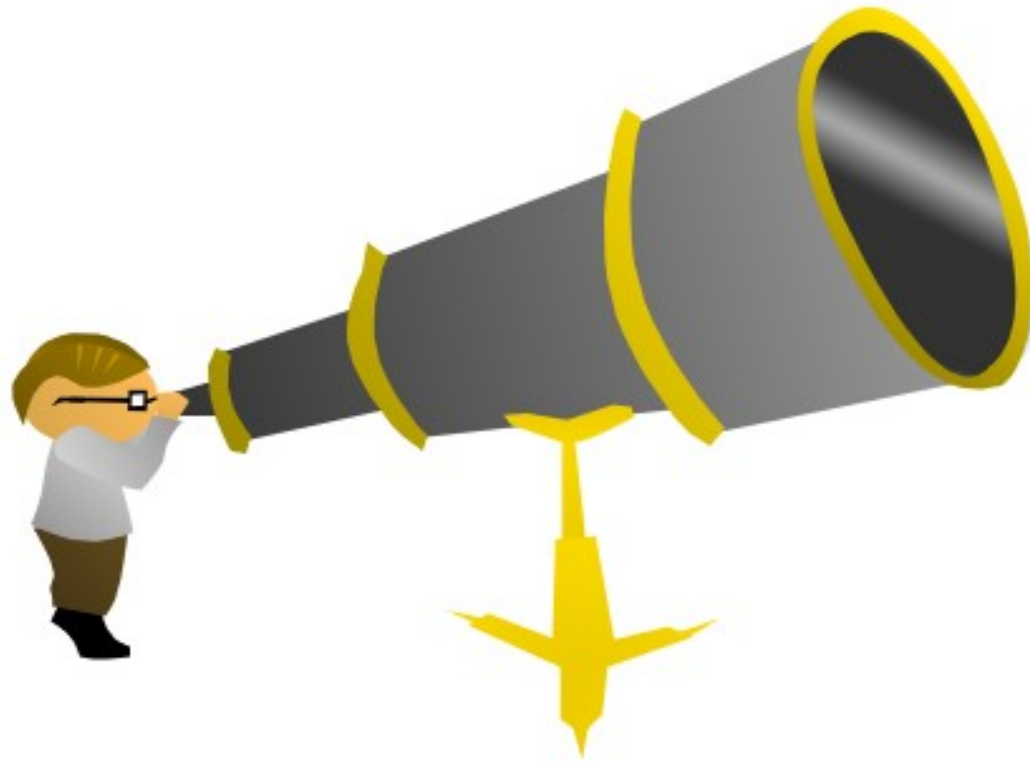
Despite the limitations of stopping at this phase, it is the most critical. Unless you can acquire and understand the surface of information, you have no chance of building depth.



Quick Tip!

Not being able to understand something instantly, isn't a sign you aren't learning holistically. My suggestion, when you encounter a tricky subject is to break it into components. Narrow down specifically what you don't understand so you can use further reading to fill the gaps.

The Explore Phase



The Explore Phase

Exploration is really the start of holistic learning. With your basic understanding that covers the surface of information, you now try to link that information to other concepts. Here you would start using models to simplify and expand your constructs. You would also broaden your understanding of not only how the information is derived, but what it is related to.

A holistic learner would take the formula I mentioned in the last phase, and explore it. Where does it come from? What do the different components of the formula represent? What elements of the formula can be altered and how does that change results? What other formulas are similar to this in function or form?

Answering these questions may take a bit of time if you aren't used to them. But once you make these methods a habit, exploration will naturally take over. Understanding won't be enough because curiosity will drive you to connect it to new material.

There are three major ways you can explore ideas: depth, lateral and vertical exploration.

Depth Exploration

Information comes from somewhere. Where does a formula come from? How was a discovery made? Why are things the way they are?

Depth exploration requires that you create links into information. Instead of just understanding a formula, you understand its proof. You understand why the proof was developed and by whom. Now, through depth exploration, your formula becomes supported underneath a foundation of other linked ideas.

The disadvantage with depth exploration is that it usually requires additional research. Some classes will provide some depth, but rarely the amount necessary to explore an idea. Depth exploration is probably the best method to lock an idea in place, but also the most time consuming.

Lateral Exploration

Information doesn't stand alone. What formulas are similar to this one in function or design? What other discoveries were made at the same time, by the same person or in the same field? What other facts surround this one?

Lateral exploration requires that you create links around information. Usually this means that you take models and link information between them. Relating one event that occurred in Ancient Greece with another event that took place in Ancient China.

Lateral exploration requires less research than depth exploration, but more creativity. It requires that you see connections that might not be present initially. Your professors might not show you all the possible similarities and comparisons one formula has to another.

Vertical Exploration

Information follows patterns, those patterns can be found in other information. Can I compare this formula to a natural event, like water flowing or driving a car? How does this discovery relate to a completely different historical event? How is this idea similar to completely different ideas.

Vertical exploration is the most difficult, but also the most creative way to learn. But, if used properly, it can be done with greater speed than the background research required in depth exploration. Vertical exploration is the biggest subject of the techniques in this book as methods such as metaphor and visceralization work well within vertical exploration.

Vertical exploration involves building those highways between constructs. This benefits your creative thinking and can give you amazing powers to understand information.

All of these three methods are equally valid. The techniques in this book can apply to any of the different types of exploration. The key is to practice and recognize what type of linking you are trying to create. Are you trying to build a foundation beneath an idea? Link an idea around similar ideas? Or create highways that bridge your idea and completely different ones?



Quick Tip!

Creating diagrams can help with vertical, lateral and depth exploration. Metaphor and visceralization both work best with vertical exploration. However, holistic learning doesn't need to be fancy, just by getting curious and asking questions you can explore an idea.

The Debug Phase



The Debug Phase

Learning is not error free. The problem with interlinking, is that a wrong connection can create false understandings. This happens when you believe you understand an idea, but the relationship it is based on is faulty.

Let's say you create a link between how corporations improve with the process of evolution. This may be a useful metaphor, but it might lack accuracy. Natural selection takes place over thousands of generations with relatively low error rates when replicating. Corporations can change completely over just a few generations with incredibly high error rates. Unless you understand places the metaphor doesn't apply, you might make reasoning errors.¹

Debugging can be seen as pruning back the holistic web. Adding exceptions and breaking connection that don't exist in reality. This isn't a perfect process, but it is necessary to seek information that will clean up the desire to find patterns.

Debugging can take many forms, from reading books from authors that oppose your points of view (and can point out errors in your reasoning) to testing your information

1. Eliezer Yudkowsky, "No Evolutions for Corporations or Nanodevices"
<http://www.overcomingbias.com/2007/11/no-evolution-fo.html>

in the world and seeing where it doesn't apply.

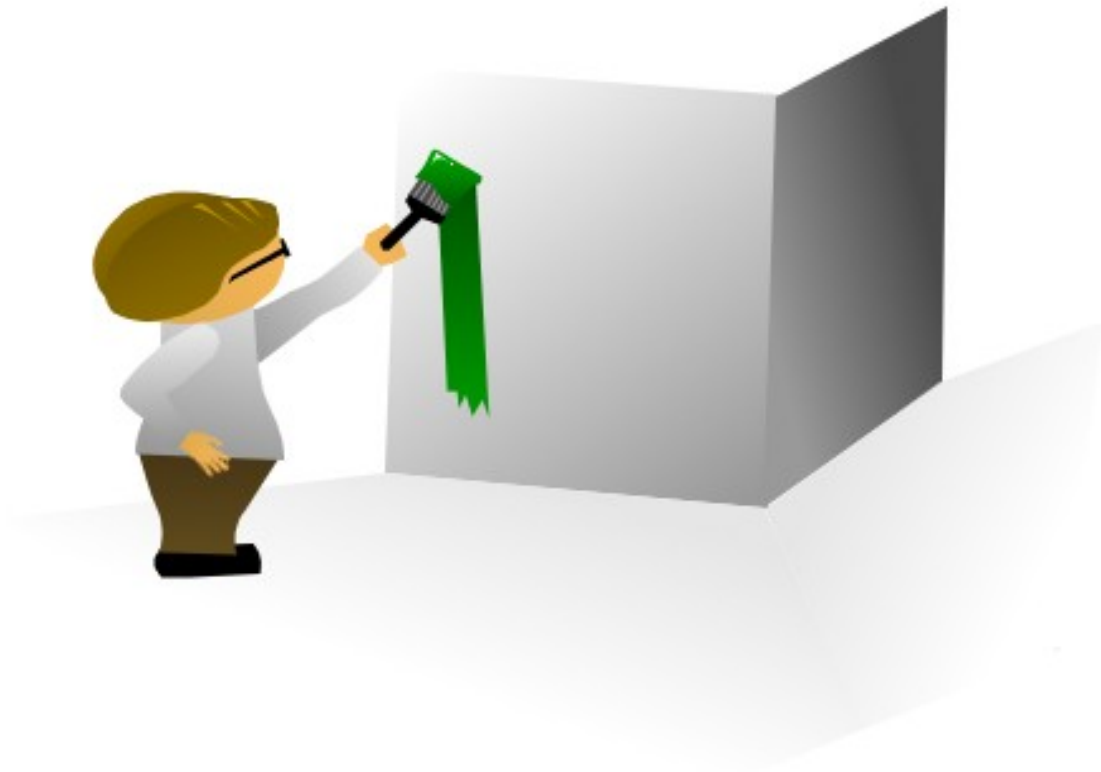
I mentioned earlier in the book how I understood what a subspace was by using a model which combined a 3-D background and red grid. Unfortunately, this is only one form of a subspace. There are many others and the grid analogy might lead to incorrect conclusions. Debugging would help me fix this glitch.



Quick Tip!

The only way to debug is to practice. You can cut down on practicing time if you form a solid foundation in the first three steps. For more suggestions on debugging, read the section on Model Debugging.

The Apply Phase



The Apply Phase

Application is where learning becomes complete. Being able to apply information makes it more likely to match the real world. You would learn tax law better if you had to apply it to your own business than just hypothetical examples. You would understand statistics better if you designed your own experiments.

Little information will actually get to reach this phase. But the more you can carry to this final stage, the more you will completely understand. Application is part practice and part creativity as you take theories and force them onto the real world.

A certain amount of all information is subconscious. That subconscious portion might not be developed if you fail to apply. Someone who has read a library of business books may understand concepts, but someone who has run a business can feel them.

Finding creative ways you can apply what you learn to your own life is the final step of holistic learning.

The Test Phase



The Test Phase

Each one of these five phases needs to be accompanied by testing. As you go through the techniques in the latter half of this e-book, you will be trying out ways of learning that might dramatically differ from your current methods. Whenever you try new methods, information might get lost or absorbed differently. Testing allows you to track what your weaknesses are, so improvements can be made.

Testing should determine what level of understanding you have for any idea. Here's the questions you need to ask yourself:

Acquire Test - Have I seen/listened to the idea before?

Understand Test - Do I get (at a surface level) what this idea means?

Explore Test - Do I understand where this idea comes from, what it is related to and what outside ideas can be connected with it?

Debug Test - Have I removed inappropriate links between this idea and others?
Have I removed false conclusions based on connections that don't actually exist?

Apply Test - Have I used this idea in my practical life?

Answering “no” to any of these points should give a clear idea of what level your understanding is with this particular idea. You might have heard the information, understood it at a basic level and connected it with other ideas – but failed to search through and delete false connections. Therefore your current level of understanding is just beyond exploration.

Testing isn’t a complicated process, but it requires a lot of self-awareness. Read over the description of the five phases again as you go through the techniques. That way you can better understand where you sit with any new idea you encounter.

Summary of Holistic Learning Sequence

All of this talk about holistic learning theory may sound a bit overwhelming. Just how am I supposed to vertically explore, create models or build constructs? At this point you may want to flip to Part II of the book. That section covers the techniques which are built on this theory.

The sequence of holistic learning isn't really a sequence at all. Instead it is a cycle that constantly jumps between steps. Understanding information goes back to Acquiring and then onto Exploration and Application before Debugging. Try to avoid seeing these steps as linear but as part of an ongoing cycle to create understandings.

Pinpointing Weaknesses in the Sequence



What are your weak links in the chain of holistic learning? Identifying weak points is a matter of regular testing. More broadly, you need to pinpoint where your chronic weaknesses are. Everyone has different strengths and weaknesses when it comes to learning. Perhaps you are great at acquiring and understanding but poor at exploring. Or perhaps you can connect ideas easily, but make too many false connections.

Finding your chronic weak points can tell you what techniques you need to master. I can't tell you what your weaknesses are. There are no concrete rules.

But here are a few guidelines you can use to see what you might need to work on:

If Your Weakness is...

...Acquiring

Poor acquisition tends to manifest itself in one of two ways:

- 1) Reading or listening takes you a long time.
- 2) You need to re-read textbooks/notes to get all of the basic information.

Some common problems that can result in poor acquiring are:

- 1) **You have bad reading/study habits.** If you're constantly reading when distracted, or don't follow reading techniques I describe in Part II of this book, you're going to take much longer to read or require multiple reads of the same material.
- 2) **You have bad note taking habits.** This can be through taking too many notes or taking too few. Too many notes prevents you from thinking while the professor talks.

Too few/poor quality notes might prevent recapturing information not properly understood the first time.

3) **You don't understand the basic terminology or the English language.** If English isn't your first language (or whatever language you're studying in) you will make errors in interpretation. If you lack the basic terminology of your field, you may get confused.

The solution to poor acquisition is to improve your study, reading and note taking habits. I have suggestions in Part II for how to do this. Acquiring isn't the most important phase of learning, so I wouldn't worry if you aren't perfect.

...Understanding

Poor understanding tends to be shown when:

- 1) You're reading but have no idea what the author is talking about.
- 2) Your notes (which are perfectly legible) don't make any sense to you.

Getting to the surface understanding can't be done through magic. My suggestion, if this is your stumbling block, is to seek as many well-written explanations as possible. Advanced techniques such as metaphor or visceralization can only work if the foundation of understanding is already there.

Even if you feel you can't understand your subjects, I've found that this is rarely the main problem. Usually most people can understand the basics of an idea. It is the cross application and linking required to remember ideas that most people stumble on.

When you don't understand, slow down and ask for different explanations. Often I'm struck by a new idea for awhile before I can fully understand it. The power I've made is being able to take the understanding, once formed, and rapidly grow that idea so that I can use it easily.

...Exploration

Poor exploration is shown by inflexibility. It happens when you get the gist of an idea, but can't relate it to other things you have learned. If you are asked to use the idea to solve a non-routine problem, you're screwed.

People who can do math problems but can't do word problems using the same techniques usually have a weakness in exploring. They understand how they've been taught to use the formula and the steps to solve the problem, but can't apply it in a different setting.

Not being able to apply ideas to different problems is a weakness in the Explore Phase, not Apply Phase. I'll make the distinction clearer by using an example:

Suppose Johnny is in a basic statistics course. He understands the basics of the formulas and remembers the examples his professor gave when solving mathematical problems.

Now Johnny is given a problem on the test which requires him to use the formula in a slightly different way than he was taught. He can't do it – the problem is in exploration.

However, if Johnny breezed through the test, chances are his exploration is strong. But, let's say, he is later confronted with a problem with a statistics problem in his life. Will he make use of the principles he learned during the course and apply them? Or will they not be able to make the jump from classroom to real-life? If he can't – that's a problem in application.

...Debugging

Poor debugging is shown by inaccurate connections. I find this is less common in the classroom, and more common in the outside world. In classes, most people do too little exploration. As a result, there is little that needs debugging, if anything.

In the outside world, superstitions aren't just the domain of religion and rituals. People have all sorts of wacky associations they've made but failed to test. Believing cheering causes a sports team to win just because cheering matches up with winning is a good example.

A sign you don't do enough debugging is if you don't regularly find yourself to be wrong on a major belief or issue. If you always think you are correct, chances are you aren't doing enough critical examination of the connections you make.

...Applying

As I mentioned with weaknesses in exploring, applying errors are caused when you can't act on ideas in the real world. Having brilliant theories is useless if you can't use them anywhere. Schools, in my opinion, fail to look at this critical aspect of learning.

People who have “book smarts” but lack common sense are usually people who fail at applying. Solving this phase is less technique and more practice. Get out there, run controlled experiments and give your ideas a test.

Identifying Your Weak Points

Shouldn't you focus on your strengths and ignore your weaknesses? Normally I'd agree with you, but this is an area where each step is crucial and can't be outsourced. If you're running a business, being the best programmer in the world isn't necessary if you can hire a programmer. As a learner, however, you can't hire someone to explore or apply ideas. You have only one brain, so you need to focus on fixing weak points in the sequence.

Once you recognize weak points, you can practice techniques to strengthen them. Initially, adopting new techniques adds a bit of time. But as you master them, they become integrated into your natural strategy for learning.

Information Structure

The process of learning is similar for most subjects. Whether it's chemistry, history or computer programming, you'll still go through the stages of acquiring, understanding, exploring, debugging and finally applying.

The problem is that all information isn't the same. You might need to remember a list of dates for a history class, and need to know how polymorphism works in a computer science class. The information in each case has a different structure. Although the process for learning is similar, it is important to take into account the structure of information.

If you eat a hamburger or a bowl of alfalfa sprouts, the process is the same. You chew the food, it is broken down in your stomach, nutrients are absorbed in your intestines and it leaves the body as waste. However a hamburger and bowl of alfalfa sprouts have very different structures. As a result you must chew, digest and absorb them in different ways to get the nutrients and remove the waste.

You can't digest information with different structures the same way. Learning must

adjust for the difference between knowing the definition of a hundred terms versus being able to argue a thesis statement.

Coming up with a solution for every type of information structure is impossible. You'll have to tweak the methods in the second half of this book towards whatever information structures you encounter. But I've come up with a few simple categories that can help you decide what structure your learning tasks fit in:

1. Arbitrary
2. Opinion
3. Process
4. Concrete
5. Abstract

These five categories define the basic information structures I've noticed in most school and real-life settings. Some information sits in-between categories: physics can often move between concrete and abstract. Biology can go between arbitrary and concrete. In fact, most topics have elements of each structure.

Arbitrary



Arbitrary Information

Arbitrary information is a set of facts, dates, definitions or rules that have no logical grouping. This type of information is often encountered in schools and less frequently in the real world. I've heard from many new medical students complain about the amount of definitions they need to memorize in anatomy class.

Methods to Use With Arbitrary Information

Your first goal with arbitrary information is to make it less arbitrary. If there is a logical pattern in the information, try to find that first. Otherwise the job of remembering and using the information becomes more difficult.

If that can't be accomplished, these methods also work well:

Linking

Pegging

Compression

Challenges with Arbitrary Information

The one benefit of arbitrary information is that it is relatively easy to understand. Knowing how a heart pumps blood requires far more intelligence than simply remembering what the different parts of a heart are called.

But this upside is also a curse. Since there is little understanding required, there is also little exploration that can be done. Therefore, this type of information is the most likely to be forgotten without sheer rote memorization. Linking, pegging and compression can help, but they can't entirely compensate for the tricky nature of this structure.



Quick Tip!

Arbitrary information is the hardest to learn holistically. If you need to remember a lot of arbitrary information for your classes, the Link, Peg and Information Compression methods can help make the process easier.

Opinion



Opinion Information

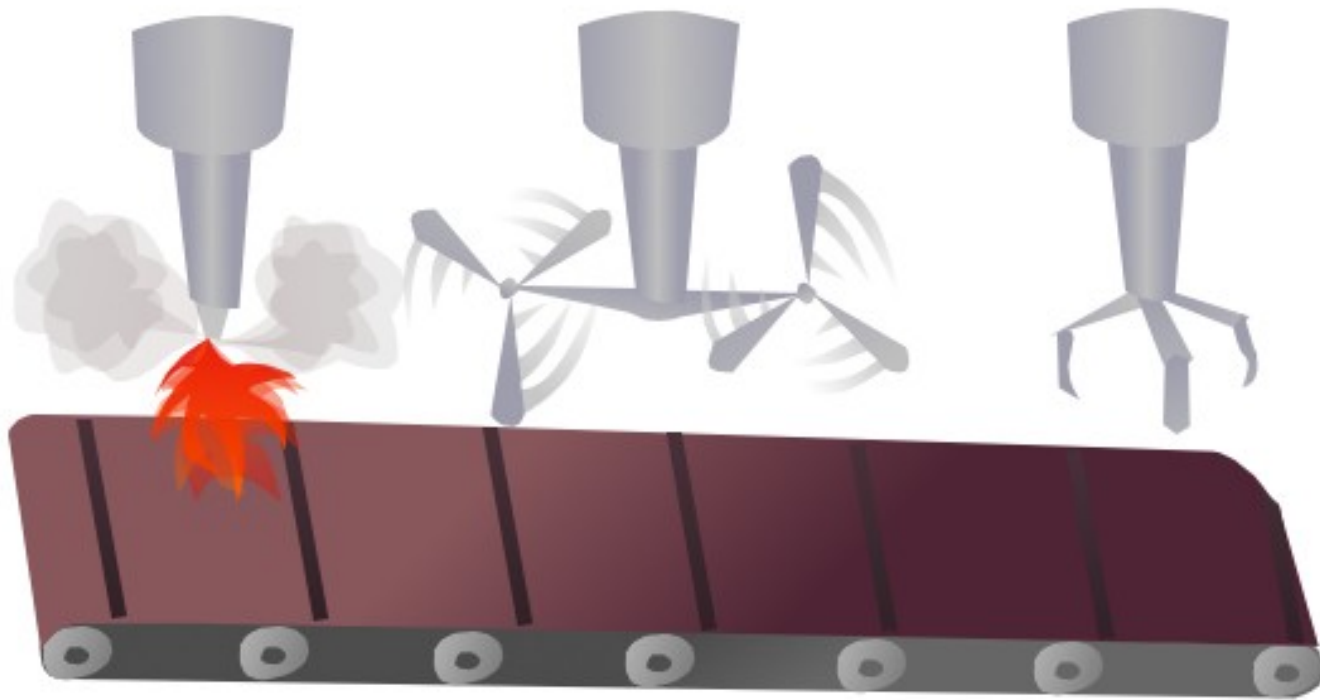
Opinion information is information you need to argue. Arguing whether Theravada Buddhism is superior to Mahayana Buddhism would require learning opinion information. This type of information is common in essays and presentations in school. This type of information is also important when making decisions where there isn't a consensus (what foods are healthy, what business opportunities are good, etc.).

Challenges With Opinion Information

With opinion information, your biggest problem is the acquiring phase. You need to examine a large volume of information and look for patterns, rather than memorize specific details. Speed reading techniques are a great asset in gathering information.

Diagraming is also a useful method with opinion information. It can help you distill your key ideas after reading.

Process



Process Information

Process information is information you need to act upon. Writing a computer program, building a house and designing a prototype are all based on process information. Practice is the most important element to learning any idea you need to act upon repeatedly, but creating the right background concepts is also crucial for saving time.

Methods to Use With Process Information

Most process information relies on having the right models. While a construct can be formed slowly with practice, having the right models can speed up learning. Here are some techniques to improve your models:

- Visceralization

- Metaphor

- Diagraming

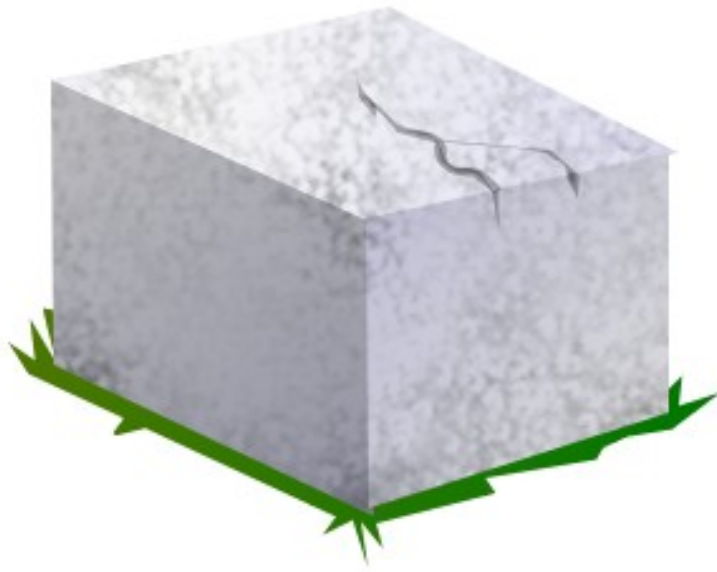
- Model Debugging

Challenges With Process Information

The biggest challenge with process information is that you need to actually put in the practice time. It doesn't matter how well you understand the concepts, until you physically perform the actions, you won't fully learn the subject material. Get the right background models and then practice the methods you've been taught.

The benefit of process information is that, if you do practice, it tends to stick with you much longer than almost any other form of information. Practice fuses the models with your construct.

Concrete



Concrete Information

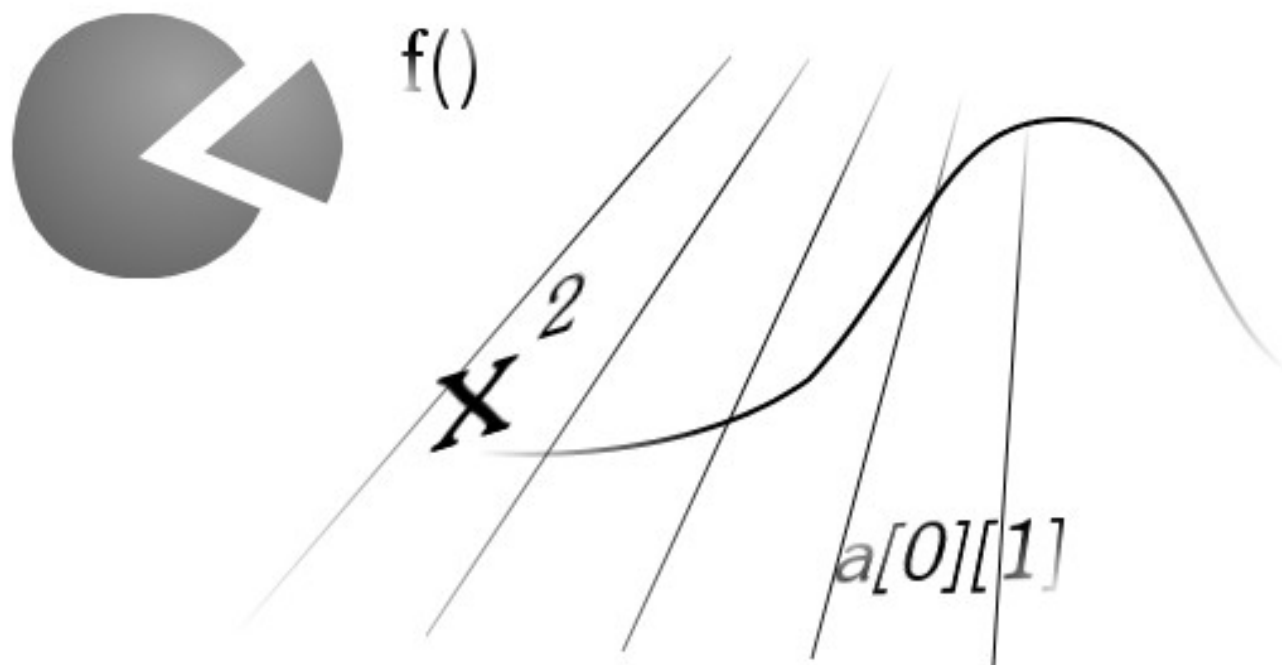
The next two categories of information, concrete and abstract, deal with the majority of information you learn in a school setting. Concrete information deals with ideas you can interact with tangibly. Most information sits on a spectrum between abstract and concrete, the difference can often be based more on the way it is presented than the content itself.

For example, biology can be a concrete subject because you have physical diagrams that represent the information. You don't need to do much work visualizing these ideas because artists and nature have done the job for you. But it can also be an abstract subject when trying to imagine enzymes, neurons or chemical reactions.

Methods to Use With Concrete Information

Most of the techniques listed in the second half of the book will work well with concrete information. Visceralization can be helpful if you want to put information into more than one of your senses. For example, you may see a diagram of a heart pumping blood, but could then translate this picture into a feeling or sound.

Abstract



Abstract Information

Abstract information is like concrete information, but it lacks the obvious connection with the senses. Mathematics, physics, psychology, computer science and chemistry are all fields that rely heavily on abstract information. Quantum physics or calculus all depend on abstract models that may not be easy to picture immediately.

Abstract information is the complete opposite of arbitrary information. Very difficult to understand but the pattern is highly logical. While arbitrary information may be incredibly shallow, abstract information can seem like diving to the bottom of an ocean.

Methods to Use With Abstract Information

When dealing with abstract information, you need to move it to a concrete level. Holistic learners usually excel in classes with arbitrary information. This is because they can understand the subject well in advance of other students who can never seem to wrap their heads around the complex ideas.

Visceralization and metaphor are essential to bring complex ideas down to a basic level. Model debugging plays prominent role because your initial attempts to model an idea may have unintended errors.

Challenges With Abstract Information

The understanding and exploring phases here can be brutal. If you feel that you don't understand a subject, slow the information stream down slightly and begin exploring the information you've been given. Since most of these courses tend to pile one idea on top of another, setting the right foundation is critical.



Quick Tip!

Abstract information is an area where holistic learning can excel over other methods. By converting information into a format that is easier to imagine, you are more likely to form links surrounding that idea. Visceralization and metaphor are both great ways to do this.

How to Use the Five Information Categories

When you need to learn a new topic, decide what major category or categories your information fits within. Deciding which category you need to learn can help you determine what techniques to use when learning and what obstacles you might face in remembering.

Weak and Strong Structures

A bridge made out of straw isn't a strong structure. A bridge with steel and careful engineering can be incredibly strong. Information structures can also be weak or

strong. You should always try to use strong structures when possible.

An arbitrary structure is the weakest structure for information. It is the hardest to learn holistically, takes the longest to learn effectively and has little value outside its specific application. If possible, try to find patterns that make arbitrary information more logical. This will help in remembering.

For example, if you have a list of definitions you need to memorize, go beyond the list. Even if the textbook offers no logical pattern, try to look for one. Maybe certain groups share a common Latin root word? Only if you can't link together logically should you try using arbitrary linking techniques such as Pegging or Compression.

Concrete and process structures are the strongest structures of information. Both of them create tangible experiences that help you link any new ideas through your senses. Creating a computer program combines concrete and process information structures. Because you can get immediate feedback through your senses about the effect of any changes, concepts quickly get wired into your brain.

You can convert a weaker structure of information into a stronger form by practicing many of the methods later in the book.

Why Your Classes are Boring

Why do you find some topics fascinating and others boring? You probably are keenly aware of which subjects interest you and which don't. But do you know why?

There are many possible reasons why you might enjoy literature over physics or computer science over accounting. You might have a natural aptitude for one subject over another. You might be able to use one subject in your daily life. One subject may even have a more interesting teacher, who can connect the abstract ideas to something you're passionate about.

The question is, are *you* in control over which subjects you like and dislike? Can you make a boring class more interesting? If you take control over how you learn, I've found you can electrify almost any subject.

Holistic learning and your interest in a class are tightly linked. The more

interesting you find the class, the easier it is to learn holistically. Do you find it easy to relate your favorite subjects, but hard to relate to boring subjects? This relationship works both ways. You can make a boring class interesting by learning more holistically.

We all have things we need to learn that initially seem boring. Maybe it's the statistics course you need to take in order to earn your degree in marketing. Maybe it's the tax law you need to learn in order to run your business. The people who can brighten these dull topics will make studying easier while learning more.

The next time you "have" to take a boring class, don't be so quick to blame the class. The problem might just be that you haven't been taught the subject in a way you can relate to. Using holistic learning methods you can connect to the ideas in a way that interests you.

When you think of computer science, do you think of dull lines of text or a secret code of the universe? When you think of accounting, do you see income statements and balance sheets, or a map charting the beating heart of a business? When you learn history do you see dates and facts, or the epic story of millions of people?

One of the best side-effects of holistic learning is that it forces you to look at subjects in a different light. Any set of ideas can be transformed from abstract principles into powerful images and stories that connect with what is most important to you.

Goals for Learning

Up until this point I've assumed your goal is to learn information completely enough to pass almost any test in a subject. But just passing tests and getting good grades isn't enough. Why do you actually want to learn this material? What usefulness can it have in your life?

Ironically, people who ask themselves questions like those are more likely to get better grades than the people who just want to pass with an A. Since they have a motivation to go beyond what is necessary to pass, they will understand the information more deeply than students who do not.

When I was starting to learn computer programming, I was fascinated by it. I went far beyond the simple lessons they taught and tried to apply the information to programs I wanted to write. I've made several decently-sized computer games as a result of this interest for programming.

When I went through an accounting class, I immediately saw ways I could apply the principles to organize my own small business. Although these methods weren't

necessary for my small business, applying some of the concepts to my records helped me both in life and in the class.

When I went through a statistics class, I found the methods for determining significance and conducting tests to be valuable for my own experiments. Now, as I conduct business and personal experiments I use these mathematical principles for getting better information.

When I went through a class teaching about vector spaces, I saw how the concepts of subspaces and vector spaces could be applied to my philosophy of life. Borrowing the concepts helped clarify ideas I had and gave me excellent metaphors for understanding.

Why so many examples? Because I want to point out that going beyond your subject doesn't need to be an issue of passion. Finding ways to use a subject (even if it currently bores you to tears) can give it new meaning.

The ultimate step in holistic learning is to take the information you have learned and give it practical value. Almost any subject can have practical value if you choose to look for it. Giving a subject value in your daily life creates a far greater connection within your mental web than any technique I can possibly describe in Part II.

Does your learning have a purpose? I don't care much about grades. I strive to do well in my courses, but the difference between a B+ and A+ doesn't really bother me.

What I do care about is having the asset of knowledge. Investing in learning can create tremendous rewards in your life if you actually use the concepts you're spending so much time learning. Learning without a practical purpose is a waste of your time.



Quick Tip!

Holistic learning can make your classes more valuable. By linking ideas together, almost any idea can become useful. Try linking some of your course material to other ideas that interest you. Every idea you learn can be used as a building block for self-improvement.

◦ Part II ◦

Holistic Learning Techniques

Holistic Learning Methods

So far you've learned the holistic learning strategy. This is my strategy for learning new concepts and how I believe learning actually works best. It is based on linking ideas together, using models, building constructs and understanding different types of information. The second half of this book is devoted to specific techniques that work within the holistic learning framework.

A good way of viewing the information of this book would be like playing chess. In order to play chess you first need to know the rules and basic objective of the game. The first half of this book could be seen as a similar set of rules and objectives when learning.

Once you understand the rules of chess, you then move onwards to specific strategies and plays to win the game. This is like the second half of the book which is devoted to different strategies I've found helpful within the rules of holistic learning.¹

1. Of course, holistic learning doesn't offer rules as precise as chess. Rather, a set of guidelines that can help.

Understanding the framework of holistic learning is the most important step in improving how you learn. Simply having techniques without the framework would make it impossible to adapt to different challenges, just as knowing chess plays without knowing the rules of the game.

All of these techniques won't work for everyone or in every circumstance. Therefore it is important to apply the holistic learning theory to isolate where your weaknesses are before deciding which techniques to use. These techniques come from a combination of my research into different learning methods and my own experiments. Therefore, you might want to modify them to suit your particular learning style.

List of Techniques

I want to stress that this is simply a starting point for improving study habits. I've read many great sources on improving learning habits with different techniques, and there are many good ideas.

Here is a list of the techniques covered in Part II:

A) Acquiring Ideas

- 1) Speed Reading
- 2) Flow-Based Notetaking

B) Linking Ideas

- 1) Metaphor
- 2) Visceralization
- 3) Diagraming

C) Handling the Arbitrary

- 1) Linking
- 2) Pegging
- 3) Information Compression

D) Extending Ideas

- 1) Practical Usage
- 2) Model Debugging
- 3) Project-Based Learning

Acquiring Ideas

The first step in learning information is to acquire it. This means actually passing the information from a textbook through your eyes, optic nerve and brain. If you never see the information in the first place, or acquire it too slowly, that will put a stop to any efforts in learning.

With acquiring information I've found two methods helpful, speed reading and flow-based notetaking.

Speed Reading



Speed Reading

If you can read faster, you can acquire more information. That's just common sense. But speed reading is more than just speed. Simply trying to read faster usually results in lost comprehension. But by practicing with different reading techniques and changing how you read, you can read faster with better understanding.

Speed reading could easily be divided up into enough techniques to fill another entire book (and indeed it has: **Breakthrough Rapid Reading** is one of my favorites). But I'm going to focus on three main sub-techniques you might want to try:

- 1) Using a pointer.
- 2) Practice reading.
- 3) Active reading.

Using a Pointer

Your eye doesn't actually stay in one place on the page as you read. Instead, it constantly makes tiny movements that slow your reading speed. Using a pointer can

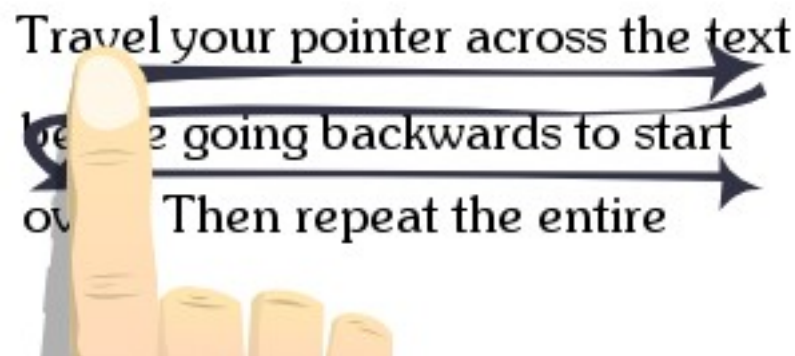
improve your reading rate simply by focusing your eye onto one part of the page—the part you want to read.

A pointer has the added benefit of allowing you to control your speed by changing the speed you move the pointer. Speed reading involves more

than just reading fast, as sometimes you will want to slow down to carefully examine one section, and skim over another that is less important.

To start using a pointer, grab a book and place your index finger right below the line you want to read. Move it across the line as you read before going to the next line. Although some speed reading experts advocate using a pointer everywhere, I tend to only use it for books. Online articles and short sections of print don't usually benefit from the extra control a pointer offers.

Using a pointer will feel uncomfortable for the first two weeks. Until you get into the habit of reading with a pointer it will probably seem slow and awkward. However, once you adapt to using your finger to scroll through the page, it will make sense. The extra control and focus it provides when reading is invaluable to boost your reading rate.



Practice Reading

Another aspect of speed reading is practice reading. Practice reading isn't the same as reading in the same way jogging on the spot isn't running. When you practice read, you expect to absorb almost none of the material you are reading. The only purpose of practice reading is to train yourself to comprehend at a faster rate.

1) You can practice read by getting a book you haven't read before. Mark a point in the book and set a timer for three minutes. Then read through the book (with your pointer) slightly faster than you can understand all of the material. When the timer goes off, mark the line you finished on.

2) Count the number of lines you read and multiply that by the average words per line on the first paragraph and divide by three. This should give you your approximate words per minute (assuming you read for three minutes).

3) This exercise can test your flexibility with improving your speed of reading.

Another exercise is directed towards improving your comprehension. Similar to the above exercise, read for three minutes using your pointer. When you are completed, start a paper writing down every major or important idea you read in the

last three minutes (without looking back at the book).

Write as many ideas as you can think of that come directly from the book. Next, go through the book again and make a second list of important ideas. Take the number of ideas you correctly wrote down after the three minute session and subtract the ones you remembered incorrectly. Then divide this number by the total meaningful ideas you gathered in your more detailed search.

$$\frac{\left(\begin{array}{c} \text{facts} \\ \text{remembered} \\ \text{correctly} \end{array} \right) - \left(\begin{array}{c} \text{facts} \\ \text{remembered} \\ \text{incorrectly} \end{array} \right)}{\text{total facts found} \\ \text{on complete search}}$$

This number represents your comprehension ability at that reading speed for that type of book. By practice reading you can aim to improve your comprehension at various words per minute, giving you greater speed and flexibility with reading.